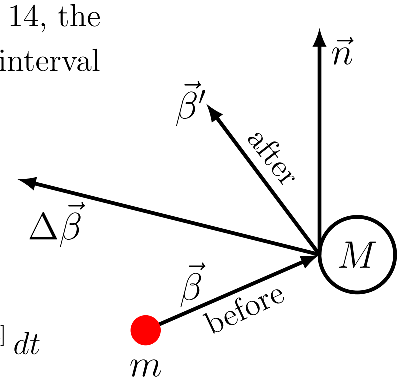
Chapter 15. Bremsstrahlung

When a charged particle passing through matter, it is scattered and losses energy as it collides with atoms. Radiation due to this atomic collision is called Bremasstrahlung (breaking radiation). Bremasstrahlung has a signature of a broad radiation spectrum.

1. Bremsstrahlung due to Hard Collision

During a hard collision, a charged particle collides with a much heavy atom ($m << M$) and it is just bounced back by switching the direction of its momentum after the collision. For an incident particle with a charge $q = Ze$, from Chapter 14, the radiation energy per unit solid angle and per unit frequency interval can be calculated from



$$\frac{d^2E}{d\Omega d\omega} = \frac{1}{\pi}|\vec{A}(\omega)|^2$$

where

$$\vec{A}(\omega) \simeq \frac{q}{\sqrt{4\pi c}} e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{\infty} \frac{\vec{n}\times[(\vec{n}-\vec{\beta}\,)\times\dot{\vec{\beta}}\,]}{(1-\vec{\beta}\cdot\vec{n}\,)^2} e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]}\, dt$$

For a hard collision, the collision occurs instantaneously without any change of the particle position. During the collision at $t = 0$, $\dot{\vec{\beta}} \neq 0$ only in a very short period of time Δt, where $\Delta t \longrightarrow 0$, and $\vec{n}$ as well as $\vec{r}$ are approximately constants. For sufficiently low frequency of the radiation, moreover, $\omega\Delta t << 1$. The radiation spectrum can then be calculated as

$$\vec{A}(\omega) = \frac{q}{\sqrt{4\pi c}} e^{i\omega(\vec{x}-\vec{r})\cdot\vec{n}/c} \int_{-\infty}^{\infty} \frac{\vec{n}\times[(\vec{n}-\vec{\beta}\,)\times\dot{\vec{\beta}}\,]}{(1-\vec{\beta}\cdot\vec{n}\,)^2}\, dt$$

With $d\vec{n}/dt \simeq 0$,

$$\frac{\vec{n}\times[(\vec{n}-\vec{\beta}\,)\times\dot{\vec{\beta}}\,]}{(1-\vec{\beta}\cdot\vec{n}\,)^2} \simeq \frac{d}{dt}\left[\frac{\vec{n}\times(\vec{n}\times\vec{\beta}\,)}{1-\vec{\beta}\cdot\vec{n}}\right]$$

and thus

$$\frac{d^2E}{d\Omega d\omega} \simeq \frac{q^2}{4\pi^2 c}\left|\left[\frac{\vec{n}\times(\vec{n}\times\vec{\beta}\,)}{1-\vec{\beta}\cdot\vec{n}}\right]_{after} - \left[\frac{\vec{n}\times(\vec{n}\times\vec{\beta}\,)}{1-\vec{\beta}\cdot\vec{n}}\right]_{before}\right|^2 \tag{1}$$

Let $\vec{\beta}\,'$ and $\vec{\beta}$ be the velocities of the incident particle before and after the collision and

$$\vec{n}\times(\vec{n}\times\vec{\beta}\,) = (\vec{n}\cdot\vec{\beta}\,)\vec{n} - \vec{\beta} = -\vec{\beta}_\perp \qquad \text{and} \qquad \vec{n}\cdot\vec{\beta} = \beta_\parallel$$

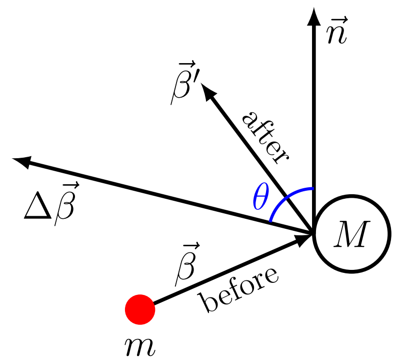
where $\vec{\beta}_\perp$ and $\beta_\parallel$ are the components of $\vec{\beta}$ perpendicular and parallel to $\vec{n}$, respectively. For a hard collision with a frequency of $\omega << 1/\Delta t$,

$$\frac{d^2E}{d\Omega d\omega} = \frac{q^2}{4\pi^2 c}\left|\frac{\vec{\beta}\,'_\perp}{1-\beta'_\parallel} - \frac{\vec{\beta}_\perp}{1-\beta_\parallel}\right|^2 \tag{2}$$

The radiation spectrum of the hard ($\Delta t \to 0$) collision is therefore independent of frequency and the radiation has a broad uniform spectrum — white noise. The reason of a broad uniform spectrum of hard collisions is because of a very short duration, impulse-like, of interaction of the collision.

a) Non-Relativisitc Case, $\beta << 1$

$$\left|\frac{\vec{n}\times(\vec{n}\times\vec{\beta}\,')}{1-\vec{\beta}\,'\cdot\vec{n}} - \frac{\vec{n}\times(\vec{n}\times\vec{\beta}\,)}{1-\vec{\beta}\cdot\vec{n}}\right|^2 \simeq |\vec{n}\times(\vec{n}\times\Delta\vec{\beta})|^2 = (\Delta\beta)^2\sin^2\theta$$



where θ is the angle between $\vec{n}$ and $\Delta\vec{\beta} = \vec{\beta}\,' - \vec{\beta}$. Then

$$\frac{d^2E}{d\Omega d\omega} = \frac{q^2}{4\pi^2 c}(\Delta\beta)^2\sin^2\theta$$

and the radiation energy per unit frequency is

$$\frac{dE}{d\omega} = \frac{q^2}{4\pi^2 c}(\Delta\beta)^2\int\sin^2\theta d\Omega = \frac{2q^2}{3\pi c}(\Delta\beta)^2 \qquad (3)$$

b) Ultra-Relativisitc Case, $\beta \sim 1$

In the case of $\gamma >> 1$, $\Delta\beta = |\vec{\beta}\,' - \vec{\beta}| << 1$. We can calculate the radiation spectrum in Eq. (2) with the lowest order of $\Delta\beta$,

$$\begin{aligned}\frac{\vec{n}\times(\vec{n}\times\vec{\beta}\,')}{1-\vec{\beta}\,'\cdot\vec{n}} - \frac{\vec{n}\times(\vec{n}\times\vec{\beta}\,)}{1-\vec{\beta}\cdot\vec{n}} &= \frac{(1-\vec{\beta}\cdot\vec{n})[\vec{n}\times(\vec{n}\times\vec{\beta}\,')] - (1-\vec{\beta}\,'\cdot\vec{n})[\vec{n}\times(\vec{n}\times\vec{\beta}\,)]}{(1-\vec{\beta}\,'\cdot\vec{n})(1-\vec{\beta}\cdot\vec{n})} \\ &\simeq \frac{\vec{n}\times(\vec{n}\times\Delta\vec{\beta}\,) + (\vec{\beta}\,'\cdot\vec{n})[\vec{n}\times(\vec{n}\times\vec{\beta}\,)] - (\vec{\beta}\cdot\vec{n})[\vec{n}\times(\vec{n}\times\vec{\beta}\,')]}{(1-\vec{\beta}\cdot\vec{n})^2} + O\left((\Delta\beta)^2\right)\end{aligned}$$

where the 1st term in the nominate is proportional to $\Delta\beta$ and the subtraction of 2nd and 3rd term is also proportional to $\Delta\beta$,

$$\begin{aligned}&(\vec{\beta}\,'\cdot\vec{n})[\vec{n}\times(\vec{n}\times\vec{\beta}\,)] - (\vec{\beta}\cdot\vec{n})[\vec{n}\times(\vec{n}\times\vec{\beta}\,')] \\ &\qquad = (\vec{\beta}\,'\cdot\vec{n})[(\vec{n}\cdot\vec{\beta}\,)\vec{n} - \vec{\beta}] - (\vec{\beta}\cdot\vec{n})[(\vec{n}\cdot\vec{\beta}\,')\vec{n} - \vec{\beta}\,'] \\ &\qquad = (\vec{\beta}\cdot\vec{n})\vec{\beta}\,' - (\vec{\beta}\,'\cdot\vec{n})\vec{\beta} = -\vec{n}\times(\vec{\beta}\times\vec{\beta}\,') = -\vec{n}\times(\vec{\beta}\times\Delta\vec{\beta})\end{aligned}$$

while the denominate is

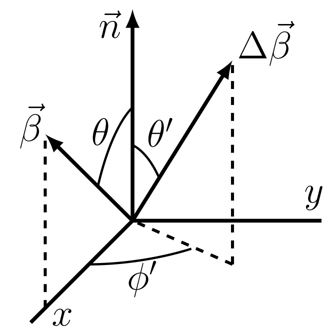
$$\begin{aligned}\frac{1}{(1-\vec{\beta}\,'\cdot\vec{n})(1-\vec{\beta}\cdot\vec{n})} &= \frac{1}{(1-\vec{\beta}\cdot\vec{n})^2 - (1-\vec{\beta}\cdot\vec{n})(\vec{n}\cdot\Delta\vec{\beta})} \\ &= \frac{1}{(1-\vec{\beta}\cdot\vec{n})^2}\left[1 - \frac{\vec{n}\cdot\Delta\vec{\beta}}{(1-\vec{\beta}\cdot\vec{n})}\right]^{-1} \simeq \frac{1}{(1-\vec{\beta}\cdot\vec{n})^2} + O(\Delta\beta)\end{aligned}$$

Therefore, up to $(\Delta\beta)^2$, we have

$$\frac{d^2E}{d\Omega d\omega} \simeq \frac{q^2}{4\pi^2 c}\left|\frac{\vec{n}\times[(\vec{n}-\vec{\beta})\times\Delta\vec{\beta}]}{(1-\vec{n}\cdot\vec{\beta})^2}\right|^2 \tag{4}$$

To go further from Eq. (4), we need to choose a coordinate. Let

$$\left\{\begin{array}{rcl} \vec{n} & = & \vec{e}_z \\ \vec{\beta} & = & \beta(\sin\theta\,\vec{e}_x + \cos\theta\,\vec{e}_z) \\ \Delta\vec{\beta} & = & \Delta\beta(\sin\theta'\cos\phi'\,\vec{e}_x + \sin\theta'\sin\phi'\,\vec{e}_y + \cos\theta'\,\vec{e}_z) \end{array}\right.$$



Then

$$\begin{array}{rcl} 1-\vec{n}\cdot\vec{\beta} & = & 1-\beta\cos\theta \\ \vec{n}-\vec{\beta} & = & -\beta\sin\theta\,\vec{e}_x + (1-\beta\cos\theta)\,\vec{e}_z \\ \vec{n}\times[(\vec{n}-\vec{\beta})\times\Delta\vec{\beta}] & = & -\left[(1-\beta\sin\theta)\Delta\beta_x + (\beta\sin\theta)\Delta\beta_z\right]\vec{e}_x - (1-\beta\cos\theta)\Delta\beta_y\,\vec{e}_y \end{array}$$

and

$$\left|\frac{\vec{n}\times[(\vec{n}-\vec{\beta})\times\Delta\vec{\beta}]}{(1-\vec{n}\cdot\vec{\beta})^2}\right|^2 = \frac{1}{(1-\beta\cos\theta)^4}\left\{[(1-\beta\sin\theta)\Delta\beta_x + (\beta\sin\theta)\Delta\beta_z]^2 + [(1-\beta\cos\theta)\Delta\beta_y]^2\right\}$$

where the first and the second square term in the bracket are of the radiation polarized in the x and y direction, respectively, which is in the transverse plane from the observer's point view. Note that $\vec{n}$ is along the z direction.

Radiation Intensity Average Over Scattering Angles (θ', ϕ')

For a charged-particle beam scattered by nuclei inside a material, the scattering angle θ' and ϕ' are randomly distributed. We average the radiation power over the scattering angles,

$$\begin{aligned} \left\langle\left|\frac{\vec{n}\times[(\vec{n}-\vec{\beta})\times\Delta\vec{\beta}]}{(1-\vec{n}\cdot\vec{\beta})^2}\right|^2\right\rangle_{\theta',\phi'} &= \frac{1}{(1-\beta\cos\theta)^4} \\ &\times \frac{1}{2\pi^2}\int_0^{2\pi} d\phi' \int_0^{\pi} d\theta'\,\left\{[(1-\beta\sin\theta)\Delta\beta_x + (\beta\sin\theta)\Delta\beta_z]^2 + [(1-\beta\cos\theta)\Delta\beta_y]^2\right\} \\ &= \frac{(\Delta\beta)^2}{4(1-\beta\cos\theta)^4}\left[(1-\beta\cos\theta)^2 + 2\beta^2\sin^2\theta + (1-\beta\cos\theta)^2\right] \end{aligned}$$

and thus

$$\begin{aligned} \left\langle\frac{d^2E}{d\Omega d\omega}\right\rangle_{\theta',\phi'} &= \frac{q^2}{8\pi^2 c}(\Delta\beta)^2\left[\frac{(1-\beta\cos\theta)^2+\beta^2\sin^2\theta}{(1-\beta\cos\theta)^4}\right] \\ &= \frac{q^2}{8\pi^2 c}(\Delta\beta)^2\left[\frac{1+\beta^2-2\beta\cos\theta}{(1-\beta\cos\theta)^4}\right] \end{aligned} \tag{5}$$

where the radiation is polarized in the transverse plane of the observation direction $\vec{n}$.

Total Radiation Energy Density

$$\left\langle \frac{dE}{d\omega} \right\rangle_{\theta',\phi'} = \frac{q^2}{8\pi^2 c} (\Delta\beta)^2 \int \frac{1+\beta^2-2\beta\cos\theta}{(1-\beta\cos\theta)^4} d\Omega$$

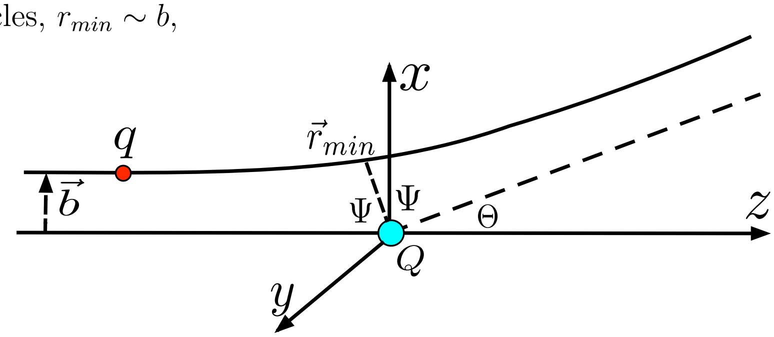
$$\overset{x=\cos\theta}{=} \frac{q^2}{4\pi c} (\Delta\beta)^2 \int_{-1}^{1} \frac{1+\beta^2-2\beta x}{(1-\beta x)^4} dx$$

$$= \frac{q^2}{6\pi c} (\Delta\beta)^2 \frac{3-\beta^2}{(1-\beta^2)^2} \overset{\beta\simeq 1}{\simeq} \frac{q^2}{3\pi c} \gamma^4 (\Delta\beta)^2 \tag{6}$$

The intensity of the radiation at $\beta \simeq 1$ is therefore γ^4 stronger than that at $\beta << 1$.

15.2 Bremsstrahlung due to Soft Collisions – Coulomb Collisions

Coulomb scattering is a "soft" collision that is the opposite limit from the "hard" collision. In Coulomb scattering, an incident charged particle interacts with a target particle *vs.* long-range Coulomb force that is proportional $1/r^2$ where r is the distance between the scattered particle and the target. As the interaction and the acceleration $\dot{\vec{\beta}}$ only fall off with $1/r^2$, the radiation lasts a long period of time along the trajectory of the scattered particle, *i.e.* $\Delta t >> 1$. Note that in the "hard collision", $\Delta t << 1$. Consider a light (m) charged particle scattered by a heavy (M) particle, *i.e.* $m << M$. Let $\vec{b}$ be the vector impact parameter of the incident particle. We also consider the case of small angle scattering with a large impact parameter, in which the total mechanical energy is much larger than the interaction potential energy. The maximal potential energy is at the closet point between the incident and target particles, $r_{min} \sim b$,



$$|\mathrm{PE}|_{max} \sim \frac{qQ}{b}$$

The mechanical energy equals the kinetic energy at $r \longrightarrow \infty$

$$\mathrm{KE} = \gamma mc^2 - mc^2 = (\gamma-1)mc^2$$

The condition of the small angle scattering is therefore

$$\mathrm{KE} >> |\mathrm{PE}|_{max} \qquad \Longrightarrow \qquad \gamma - 1 >> \frac{qQ}{mc^2 b}$$

General Formalism for Solving Scattering Problem

To calculate the radiation due to scattering of a charged particle, we need the acceleration as well as the trajectory of the scattered particle. For an arbitrary (small or large angle) scattering, the motion of the scattered particle with $m << M$ needs to be salved from a central force problem with a Coulomb force,

$$\frac{d\vec{p}}{dt} = \frac{qQ}{r^3}\,\vec{r} \tag{7}$$

For example, for an non-relativistic particle, this scattering problem can be salved with integrals (from your mechanics textbook) of

$$\begin{cases} dt &= \dfrac{\sqrt{M}\,dr}{[2(E-V(r))]^{1/2}} \\ d\psi &= \dfrac{M}{mr^2}dt \end{cases} \qquad \Longrightarrow \qquad d\psi = \frac{\sqrt{M}}{[2(E-V(r))]^{1/2}}\,dr$$

where $E,\ M = mv_0 b$, and v_0 are the mechanical energy, the constant angular momentum, and the initial speed of the incident particle, respectively, and $V(r)$ is the effective potential

$$V(r) = \frac{M^2}{2mr^2} - \frac{qQ}{r}$$

Scattering in Small Angle Approximation

For a small angle scattering,

$$\gamma - 1 >> \frac{qQ}{mc^2 b} \ \sim \ \epsilon$$

where ϵ is the small scale of the interaction between the particles during the small-angle scattering, the particle is almost traveling along a straight line with an approximately constant velocity $\vec{\beta}$. To calculate of the radiation in the case of small angle scattering, we will use the lowest order solution of the particle trajectory and acceleration. For convenience, we choose $t = 0$ when the scattered particle is at the closet point to the target, *i.e.* $\vec{r}(0) \simeq \vec{b}$, the lowest-order, zeroth-order, solution of the particle trajectory is then

$$\vec{r}(t) \simeq \vec{b} + c\vec{\beta}\,t + O\left(\epsilon\right)$$

where $\vec{b}$ and $\vec{\beta}$ are perpendicular to each other. With the zeroth-order solution of the trajectory, the 1st-order approximation of the acceleration of the scattered particle can be calculation as

$$\frac{d\vec{p}}{dt} = \frac{qQ}{r^3}\,\vec{r} = \frac{qQ}{\left[(\beta ct)^2 + b^2\right]^{3/2}}\left(\vec{\beta}ct + \vec{b}\right) + O\left(\epsilon^2\right) \tag{8}$$

With

$$\frac{d\vec{p}}{dt} = \frac{d}{dt}\left(\gamma mc\vec{\beta}\right) = mc\gamma\left[\dot{\vec{\beta}} + \gamma^2\left(\vec{\beta}\cdot\dot{\vec{\beta}}\right)\vec{\beta}\right]$$

Eq. (8) can be written as

$$\left[\dot{\vec{\beta}} + \gamma^2 \left(\vec{\beta}\cdot\dot{\vec{\beta}}\right)\vec{\beta}\right] = \alpha\left(\vec{\beta}ct + \vec{b}\right) + O\left(\epsilon^2\right) \qquad \text{where} \quad \alpha = \frac{qQ}{mc\gamma\left[(\beta ct)^2 + b^2\right]^{3/2}}$$

To solve $\dot{\vec{\beta}}$ from this equation, dot-producting $\vec{\beta}$ and $\vec{b}$, respectively, on the both sides of the equation yield

$$\left\{\begin{array}{rcl} \left(\vec{\beta}\cdot\dot{\vec{\beta}}\right)(1+\gamma^2\beta^2) & = & \alpha\beta^2 ct \\ \vec{b}\cdot\dot{\vec{\beta}} & = & \alpha b^2 \end{array}\right. \qquad \Longrightarrow \qquad \left\{\begin{array}{rcl} \dot{\vec{\beta}}\cdot\left(\dfrac{\vec{\beta}}{\beta}\right) & = & \alpha\dfrac{\beta ct}{\gamma^2} \\ \dot{\vec{\beta}}\cdot\left(\dfrac{\vec{b}}{b}\right) & = & \alpha b \end{array}\right.$$

where $(1+\gamma^2\beta^2) = \gamma^2$. Since $\vec{\beta}\perp\vec{b}$,

$$\dot{\vec{\beta}} = \alpha\frac{\beta ct}{\gamma^2}\left(\frac{\vec{\beta}}{\beta}\right) + \alpha b\left(\frac{\vec{b}}{b}\right) = \frac{qQ}{\gamma^3 mc}\,\frac{\vec{b}\,\gamma^2 + \vec{\beta}\,ct}{\left[(\beta ct)^2 + b^2\right]^{3/2}} + O\left(\epsilon^2\right) \tag{9}$$

Radiation due to Small-Angle Scattering

The spectrum of the radiation due to the small-angle scattering is calculated from the Fourier transformation of the radiation field,

$$\vec{A}(\omega) = \frac{q}{\sqrt{4\pi c}}\,e^{i\omega(\vec{x}\cdot\vec{n})/c}\int_{-\infty}^{\infty}\frac{\vec{n}\times[(\vec{n}-\vec{\beta}\,)\times\dot{\vec{\beta}}\,]}{(1-\vec{\beta}\cdot\vec{n}\,)^2}\,e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]}\,dt \tag{10}$$

where $\vec{n}$ is treated as a constant because the observation distance from the scattering is much larger than the interaction range of the trajectory of the scattered particle. For a 1st-order calculation of Eq. (10), $\vec{\beta}$ can be approximated as a constant because $\dot{\vec{\beta}}$ in Eq. (9) is already the 1st-order small. Therefore,

$$\left\{\begin{array}{rcl} \vec{\beta} & = & constant + O(\epsilon) \\ \dot{\vec{\beta}} & = & O(\epsilon) \\ \vec{r}(t) & = & \vec{r}(0) + c\vec{\beta}\,t + O(\epsilon) \\ e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]} & = & e^{i\omega\left[t\left(1-\vec{n}\cdot\vec{\beta}\,\right)+\vec{n}\cdot\vec{r}(0)/c\right]}\,e^{iO(\epsilon)} \\ & = & e^{i\omega t\left(1-\vec{n}\cdot\vec{\beta}\,\right)}\,e^{i(constant)} + O(\epsilon) \end{array}\right.$$

and the Fourier transformation of the radiation field can be simplified into

$$\vec{A}(\omega) = \frac{q}{\sqrt{4\pi c}}\,e^{i\omega(\vec{x}\cdot\vec{n})/c}\frac{\vec{n}}{(1-\vec{\beta}\cdot\vec{n}\,)^2}\times\left[(\vec{n}-\vec{\beta}\,)\times\int_{-\infty}^{\infty}\dot{\vec{\beta}}\,e^{i\omega t(1-\vec{n}\cdot\vec{\beta}\,)}\,dt\right] + O\left(\epsilon^2\right) \tag{11}$$

where the constant phase in the exponential is not important and neglected. Substituting $\dot{\vec{\beta}}$ in Eq. (9), the integral for $\vec{A}(\omega)$ becomes

$$\int_{-\infty}^{\infty} \dot{\vec{\beta}}\, e^{i\omega t(1-\vec{n}\cdot\vec{\beta}\,)}\, dt \;=\; \frac{qQ}{\gamma^3 mc} \int_{-\infty}^{\infty} \frac{\vec{b}\,\gamma^2 + \vec{\beta}\,ct}{(\beta^2c^2t^2+b^2)^{3/2}}\, e^{i\omega t(1-\vec{n}\cdot\vec{\beta}\,)}\, dt \tag{12}$$

Let

$$\tau = \frac{b}{\beta c}(1-\vec{n}\cdot\vec{\beta}\,) \qquad \text{and} \qquad t = (b/\beta c)\,\xi$$

The two terms in the integral in Eq. (12) can be calculated as

$$\int_{-\infty}^{\infty} \frac{1}{(\beta^2c^2t^2+b^2)^{3/2}}\, e^{i\omega t(1-\vec{n}\cdot\vec{\beta}\,)}\, dt \;=\; \frac{1}{\beta c\, b^2}\int_{-\infty}^{\infty} \frac{e^{i(\omega\tau)\xi}}{(\xi^2+1)^{3/2}}\, d\xi \;=\; \frac{2\omega\tau}{\beta c\, b^2}\, K_1(\omega\tau)$$

$$\int_{-\infty}^{\infty} \frac{t}{(\beta^2c^2t^2+b^2)^{3/2}}\, e^{i\omega t(1-\vec{n}\cdot\vec{\beta}\,)}\, dt \;=\; \frac{1}{\beta^2c^2 b}\int_{-\infty}^{\infty} \frac{\xi\, e^{i(\omega\tau)\xi}}{(\xi^2+1)^{3/2}}\, d\xi \;=\; i\,\frac{2\omega\tau}{\beta^2c^2 b}\, K_0(\omega\tau)$$

where $K_0(\omega\tau)$ and $K_1(\omega\tau)$ are the modified Bessel functions,

$$\int_{-\infty}^{\infty} \frac{e^{i(\omega\tau)\xi}}{(\xi^2+1)^{3/2}}\, d\xi = 2(\omega\tau)K_1(\omega\tau) \qquad \text{and} \qquad \int_{-\infty}^{\infty} \frac{\xi\, e^{i(\omega\tau)\xi}}{(\xi^2+1)^{3/2}}\, d\xi = i2(\omega\tau)K_0(\omega\tau)$$

The integral for $\vec{A}(\omega)$ in Eq. (11) can therefore be written as

$$\int_{-\infty}^{\infty} \dot{\vec{\beta}}\, e^{i\omega t(1-\vec{n}\cdot\vec{\beta}\,)}\, dt \;=\; \frac{2qQ}{\gamma^3 mc^2\beta}\,\frac{\omega\tau}{b^2}\left[\vec{b}\,\gamma^2 K_1(\omega\tau) + i\vec{\beta}\left(\frac{b}{\beta}\right) K_0(\omega\tau)\right]$$

$$=\; \frac{2qQ}{\gamma^3 mc^2\beta}\,\frac{\omega\tau}{b}\left[\hat{b}\,\gamma^2 K_1(\omega\tau) + i\hat{\beta} K_0(\omega\tau)\right]$$

where

$$\omega\tau = \frac{b\omega}{\beta c}(1-\vec{\beta}\cdot\vec{n}\,)\,, \qquad \hat{b} = \vec{b}/b\,, \qquad \hat{\beta} = \vec{\beta}/\beta$$

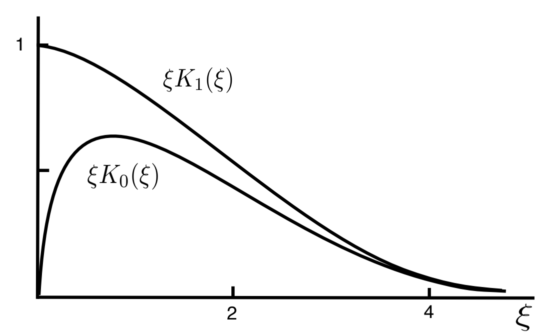
Note that

$$\lim_{\xi\to 0}\xi K_0(\xi) = 0\,, \qquad \lim_{\xi\to\infty}\xi K_0(\xi) = 0$$

$$\lim_{\xi\to 0}\xi K_1(\xi) = 1\,, \qquad \lim_{\xi\to\infty}\xi K_1(\xi) = 0$$

where $\xi = \omega\tau$. Thus,

$$\vec{A}(\omega) \longrightarrow 0 \qquad \text{when} \quad \omega\tau >> 1$$



Therefore, the Bremsstrahlung spectrum falls off exponentially at large frequency of

$$\omega >> \frac{1}{\tau} = \frac{\beta c/b}{(1-\vec{\beta}\cdot\vec{n}\,)}$$

Note that $1/(1-\vec{\beta}\cdot\vec{n}\,) >> 1$ when $\vec{n}$ is in parallel with $\vec{\beta}$ approximately and $\beta \simeq 1$.

Radiation Energy

$$
\begin{aligned}
\frac{d^2E}{d\Omega d\omega} &= \frac{1}{\pi}\left|\vec{A}(\omega)\right|^2 \\
&= \frac{q^2}{4\pi^2 c(1-\vec{\beta}\cdot\vec{n}\,)^4}\left(\frac{2qQ}{\gamma^3 mc^2\beta}\,\frac{\omega\tau}{b}\right)^2 \\
&\quad\times\left\{\left|\vec{n}\times\left[(\vec{n}-\vec{\beta}\,)\times\hat{b}\right]\right|^2\gamma^4 K_1^2(\omega\tau) + \left|\vec{n}\times\left[(\vec{n}-\vec{\beta}\,)\times\hat{\beta}\right]\right|^2 K_0^2(\omega\tau)\right\}
\end{aligned}
\tag{13}
$$

Since $\hat{\beta}\cdot\hat{b}\simeq 0\,,$

$$
\begin{aligned}
\vec{n}\times\left[(\vec{n}-\vec{\beta}\,)\times\hat{b}\right] &= (\vec{n}-\vec{\beta}\,)(\vec{n}\cdot\hat{b}) - \hat{b}\left[\vec{n}\cdot(\vec{n}-\vec{\beta}\,)\right] \\
&= (\vec{n}-\vec{\beta}\,)(\vec{n}\cdot\hat{b}) - \hat{b}(1-\vec{n}\cdot\vec{\beta}\,) \\
\left|\vec{n}\times\left[(\vec{n}-\vec{\beta}\,)\times\hat{b}\right]\right|^2 &= \left[(\vec{n}-\vec{\beta}\,)(\vec{n}\cdot\hat{b}) - \hat{b}(1-\vec{n}\cdot\vec{\beta}\,)\right]\cdot\left[(\vec{n}-\vec{\beta}\,)(\vec{n}\cdot\hat{b}) - \hat{b}(1-\vec{n}\cdot\vec{\beta}\,)\right] \\
&= |\vec{n}-\vec{\beta}\,|^2(\vec{n}\cdot\hat{b})^2 + (1-\vec{n}\cdot\vec{\beta}\,)^2 - 2\left[(\vec{n}-\vec{\beta}\,)\cdot\hat{b}\right](1-\vec{n}\cdot\vec{\beta}\,)(\vec{n}\cdot\hat{b}) \\
&= (1+\beta^2-2\vec{n}\cdot\vec{\beta}\,)(\vec{n}\cdot\hat{b})^2 + (1-\vec{n}\cdot\vec{\beta}\,)^2 - 2(1-\vec{n}\cdot\vec{\beta}\,)(\vec{n}\cdot\hat{b})^2 \\
&= (\beta^2-1)(\vec{n}\cdot\hat{b})^2 + (1-\vec{n}\cdot\vec{\beta}\,)^2 \\
\vec{n}\times\left[(\vec{n}-\vec{\beta}\,)\times\hat{\beta}\right] &= \vec{n}\times(\vec{n}\times\hat{\beta}\,) \;=\; \vec{n}(\vec{n}\cdot\hat{\beta}\,) - \hat{\beta} \\
\left|\vec{n}\times\left[(\vec{n}-\vec{\beta}\,)\times\hat{\beta}\right]\right|^2 &= \left[\vec{n}(\vec{n}\cdot\hat{\beta}\,)-\hat{\beta}\right]\cdot\left[\vec{n}(\vec{n}\cdot\hat{\beta}\,)-\hat{\beta}\right] \\
&= (\vec{n}\cdot\hat{\beta}\,)^2 + 1 - 2(\vec{n}\cdot\hat{\beta}\,)^2 \;=\; 1-(\vec{n}\cdot\hat{\beta}\,)^2
\end{aligned}
$$

and

$$
\begin{aligned}
\frac{d^2E}{d\Omega d\omega} &= \frac{1}{c}\left[\frac{Qr_0\omega}{\pi\gamma^3\beta^2 c(1-\vec{\beta}\cdot\vec{n}\,)}\right]^2 \\
&\quad\times\left\{\left[\gamma^2(1-\vec{n}\cdot\vec{\beta}\,)^2 - (\vec{n}\cdot\hat{b})^2\right]\gamma^2 K_1^2(\omega\tau) + \left[1-(\vec{n}\cdot\hat{\beta}\,)^2\right]K_0^2(\omega\tau)\right\}
\end{aligned}
$$

where $r_0 = q^2/(mc^2)$ is the classical radius of the incident particle. For electron, $q = e$ and $r_0 = 2.8\times 10^{-15}\,$m. With the coordinate of

$$
\hat{\beta} = \vec{e}_z\,,\qquad \hat{b} = \vec{e}_x\,,\qquad \text{and}\qquad \vec{n} = \sin\theta\cos\phi\,\vec{e}_x + \sin\theta\sin\phi\,\vec{e}_y + \cos\theta\,\vec{e}_z
$$

where θ is the angle between the $\vec{n}$ and $\vec{\beta}$.

$$
\begin{aligned}
\frac{d^2E}{d\Omega d\omega} &= \frac{1}{c}\left[\frac{Qr_0\omega}{\pi\gamma^3\beta^2 c(1-\beta\cos\theta)}\right]^2 \\
&\quad\times\left\{\left[\gamma^2(1-\beta\cos\theta)^2 - \sin^2\theta\cos^2\phi\right]\gamma^2 K_1^2(\omega\tau) + K_0^2(\omega\tau)\,\sin^2\theta\right\}
\end{aligned}
\tag{14}
$$

where

$$
\tau = \frac{b}{\beta c}(1-\beta\cos\theta)
$$

a) Non-Relativistic Case, $\beta << 1$

$$\frac{d^2E}{d\Omega d\omega} = \frac{1}{c}\left(\frac{Qr_0\omega}{\pi\beta^2 c}\right)^2\left[K_1^2(\omega\tau)(1-\sin^2\theta\cos^2\phi)+K_0^2(\omega\tau)\sin^2\theta\right]$$

where $\tau = b/(\beta c)$ is no longer a function of θ. The frequency spectrum of total radiation energy is then

$$\begin{aligned}\frac{dE}{d\omega} &= \frac{1}{c}\left(\frac{Qr_0\omega}{\pi\beta^2 c}\right)^2\int d\Omega\left[K_1^2(\omega\tau)(1-\sin^2\theta\cos^2\phi)+K_0^2(\omega\tau)\sin^2\theta\right]\\ &= \frac{8\pi}{3c}\left(\frac{Qr_0\omega}{\pi\beta^2 c}\right)^2\left[K_1^2(\omega\tau)+K_0^2(\omega\tau)\right] \qquad (15)\end{aligned}$$

For an uniform incident beam of charged particles with beam radius b_{max}, the total radiation energy can be calculated by the integration of $(dI/d\omega)\,(b\,db\,d\phi)$ over the cross section of the beam, where $(b\,db\,d\phi)$ is the area element on annulus on the cross section of the incident beam,

$$\frac{dE_{beam}}{d\omega} = \int_0^{b_{max}} b\,db\int_0^{2\pi} d\phi\,\frac{dE}{d\omega} = \frac{16}{3c}\left(\frac{Qr_0}{\beta}\right)^2\int_0^{x_{max}}\left[K_1^2(x)+K_0^2(x)\right]x dx \qquad (16)$$

where $x = \omega\tau = \omega b/(\beta c)$ and $x_{max} = \omega b_{max}/(\beta c)$. This integral of Bessel functions will be discussed in the next section. Note that the Bremsstrahlung radiation of a beam with $\beta << 1$ is independent of ω up to a maximal radiation frequency of $\omega_{max} \simeq \beta c/b_{min}$, *i.e.* it is a broad spectrum radiation, where b_{min} is the minimal impact parameter for a "soft" Coulomb collision collision.

b) Ultra-Relativistic Case with $\gamma >> 1$

$$1-\beta^2 = \frac{1}{\gamma^2} \qquad \Longrightarrow \qquad \beta = \sqrt{1-\frac{1}{\gamma^2}} \simeq 1-\frac{1}{2\gamma^2}$$

Since $d^2E/d\Omega d\omega$ in Eq. (14) contains a small denominator,

$$\frac{1}{(1-\beta\cos\theta)^2} \simeq \frac{1}{(1-\cos\theta)^2}$$

that diverges at $\theta \sim 0$, the radiation is confined in a narrow forward cone with an opening angle of $\theta << 1$. One can therefore use the small θ expansion, $\sin\theta \simeq \theta$ and $\cos\theta \simeq 1-\theta^2/2$, in Eq. (14) for the radiation energy, *i.e.*

$$(1-\beta\cos\theta) \simeq 1-\left(1-\frac{1}{2\gamma^2}\right)\left(1-\frac{1}{2}\theta^2\right) \simeq \frac{1}{2\gamma^2}(1+\gamma^2\theta^2) \qquad (17)$$

and the coefficient of the $K_1^2(\omega\tau)$-term in Eq. (14) can be written as

$$\left[\gamma^2(1-\cos\theta)^2-\sin^2\theta\cos^2\phi\right] \simeq \frac{1}{4\gamma^2}\left(1+\gamma^4\theta^4-2\gamma^2\theta^2\cos 2\phi\right)$$

The radiation spectrum of a relativistic particle in the soft collision is then

$$\begin{aligned}
\frac{d^2E}{d\Omega d\omega} &= \frac{1}{c}\left[\frac{Qr_0\omega}{\pi\gamma^3\beta^2 c(1-\beta\cos\theta)}\right]^2 \\
&\quad \times \left\{\left[\gamma^2(1-\beta\cos\theta)^2 - \sin^2\theta\cos^2\phi\right]\gamma^2 K_1^2(\omega\tau) + K_0^2(\omega\tau)\,\sin^2\theta\right\} \\
&\simeq \frac{4}{c}\left[\frac{Qr_0\omega}{\pi c\gamma(1+\gamma^2\theta^2)}\right]^2 \left[\frac{1}{4\gamma^2}\left(1+\gamma^4\theta^4 - 2\gamma^2\theta^2\cos 2\phi\right)K_1^2(\omega\tau) + \theta^2 K_0^2(\omega\tau)\right] \\
&\simeq \frac{1}{c}\left[\frac{Qr_0\omega}{\pi c\gamma^2(1+\gamma^2\theta^2)}\right]^2 \left(1+\gamma^4\theta^4 - 2\gamma^2\theta^2\cos 2\phi\right)\,K_1^2(\omega\tau) \\
&= \frac{1}{c}\left[\frac{2Qr_0}{\pi b(1+\gamma^2\theta^2)^2}\right]^2 \left(1+\gamma^4\theta^4 - 2\gamma^2\theta^2\cos 2\phi\right)\,\left[(\omega\tau)K_1(\omega\tau)\right]^2 \qquad (18)
\end{aligned}$$

where the coefficient of the $K_0^2(\omega\tau)$-term is $\theta^2 << 1$ and can be neglected, and

$$\tau = \frac{b}{\beta c}(1-\beta\cos\theta) \simeq \frac{b}{2c\gamma^2}(1+\gamma^2\theta^2) \qquad (19)$$

The total radiation energy per solid angle is then

$$\begin{aligned}
\frac{dE}{d\Omega} &= \int_0^\infty \frac{d^2E}{d\Omega d\omega}d\omega \\
&\overset{x=\omega\tau}{=} \frac{1}{\tau c}\left[\frac{2Qr_0}{\pi b(1+\gamma^2\theta^2)^2}\right]^2 \left(1+\gamma^4\theta^4 - 2\gamma^2\theta^2\cos 2\phi\right)\int \left[xK_1(x)\right]^2 dx \\
&= \frac{8\gamma^2 Q^2 r_0^2}{\pi^2 b^3\,(1+\gamma^2\theta^2)^5}\left(1+\gamma^4\theta^4 - 2\gamma^2\theta^2\cos 2\phi\right)\int_0^\infty \left[xK_1(x)\right]^2 dx \qquad (20)
\end{aligned}$$

where the integral of $\int_0^\infty [xK_1(x)]^2\,dx < 4$ is a finite constant. Since

$$\frac{dE}{d\Omega} \sim \left(\frac{Qq}{b}\right)\left(\frac{\gamma r_0}{b}\right)^2 << \gamma mc^2$$

where $(Qq/b) << \gamma mc^2$ is the maximal potential energy of the incident particle, which is much smaller than the kinetic energy of the particle, and $r_0 << 10^{-13}m << b$. The energy loss of the incident particle due to the the radiation is therefore much smaller than the kinetic energy of the particle when $\gamma >> 1$.

Radiation Energy From a Beam of Charged Particles

The total energy radiated from a uniform beam of charged particles passing through matter can be calculated by the integration over the cross section of incident beam with a beam radius b_{max},

$$\begin{aligned}
\frac{d^2 E_{beam}}{d\Omega d\omega} &= \int_0^{b_{max}} b\,db \int_0^{2\pi} d\phi \frac{d^2 E}{d\Omega d\omega} \\
&= \frac{1}{c}\left[\frac{Q r_0 \omega}{\pi c\gamma(1+\gamma^2\theta^2)}\right]^2 \int_0^{b_{max}} b\,db \int_0^{2\pi} d\phi \left(1+\gamma^4\theta^4 - 2\gamma^2\theta^2\cos 2\phi\right) K_1^2(\omega\tau) \\
&\overset{x=\omega\tau}{=} \frac{8Q^2 r_0^2\gamma^2(1+\gamma^4\theta^4)}{\pi c(1+\gamma^2\theta^2)^4} \int_0^{b_{max}} [xK_1(x)]^2 \frac{dx}{x}
\end{aligned} \tag{21}$$

where

$$x = \omega\tau = \frac{\omega(1+\gamma^2\theta^2)}{2c\gamma^2} b$$

Since

$$\lim_{x\to 0} xK_1(x) = 1$$

this integral diverges at the lower bound. This divergency is, however, non-physical because when $x \sim b \to 0$, the condition of small angle scattering is invalid. Recall that the necessary condition for a small angle scattering is

$$\gamma - 1 >> \frac{qQ}{mc^2 b} \tag{22}$$

The smallest b of a particle could have in a beam that satisfies the small-angle approximation is therefore

$$\frac{qQ}{mc^2 b_{min}} \sim 1 \qquad \text{or} \qquad b_{min} \sim \frac{qQ}{mc^2} \tag{23}$$

The integral in Eq. (21) should thus have a cutoff at lower bound at this b_{min}. There is another quantum limit for the lower bound of b. Since the angular momentum of the incident particle is

$$p_\theta = bp = b\gamma mv \tag{24}$$

which has to be larger than $\hbar$, *i.e.*

$$b_{min} > \frac{\hbar}{\gamma mv} \simeq \frac{\hbar}{\gamma mc} \tag{25}$$

Therefore, the lower bound of the integral in Eq. (21) has to be

$$b_{min} = \text{Max}\left\{ \frac{qQ}{mc^2}\, ,\ \frac{\hbar}{\gamma mc} \right\} \tag{26}$$

At the other end of the integral in Eq. (21), for $x > 1$, $xK_1(x) \sim e^{-x}$ and thus

$$\int^{\infty} \frac{dx}{x}[xK_1(x)]^2 \longrightarrow 0$$

We can also use a cutoff at the upper bound of the integral with

$$x_{max} \simeq 1$$

or

$$b_{max} = \frac{\beta c x_{max}}{\omega(1-\beta\cos\theta)} \simeq \frac{2c\gamma^2}{\omega(1+\gamma^2\theta^2)} \tag{27}$$

Using the cutoffs at both the lower and upper bound, the integral over the impact parameter in Eq. (21) can be approximated as

$$\int_0^\infty \frac{[xK_1(x)]^2}{x}\,dx \simeq \int_{x_{min}}^{x_{max}} \frac{[xK_1(x)]^2}{x}\,dx \simeq \int_{x_{min}}^{x_{max}} \frac{dx}{x} = \ln\left(\frac{b_{max}}{b_{min}}\right)$$

where $xK_1(x) \simeq 1$ for $x_{min} < x < x_{max}$. The total energy radiated from a beam due to the soft scattering is thus

$$\frac{d^2E_{beam}}{d\Omega d\omega} = \int_0^\infty b\,db \int_0^{2\pi} d\phi \frac{d^2E}{d\Omega d\omega} \simeq \frac{8Q^2r_0^2\gamma^2(1+\gamma^4\theta^4)}{\pi c(1+\gamma^2\theta^2)^4} \ln\left(\frac{b_{max}}{b_{min}}\right) \tag{28}$$

Cutoff Frequency of the Soft-Scattering Radiation

When $x = \omega\tau > 1$, $xK_1(x) \sim 0$ and, therefore, there is very little radiation energy for $\omega\tau > 1$. The cutoff frequency of tje radiation is then

$$\omega_{max}\tau = \frac{b\omega_{max}}{\beta c}(1-\beta\cos\theta) \sim 1 \qquad \Longrightarrow \qquad \omega_{max} \sim \frac{2\gamma^2\beta c}{b(1+\gamma^2\theta^2)} \sim \frac{\gamma^2 c}{b_{min}} \tag{29}$$

where $(1-\beta\cos\theta) \simeq (1+\gamma^2\theta^2)/2\gamma^2$ for $\gamma >> 1$ and $\theta << 1$.

Extra HW 1. This is a Classical Mechanics problem – the scattering between two particles.

For an incident and a target particle, with an interaction potential energy $V(r) = -K/r$, where K is a constant and r is the distance between the two particles, the Hamiltonian is

$$H = \frac{1}{2m_1}(\vec{p}_1 \cdot \vec{p}_1) + \frac{1}{2m_2}(\vec{p}_2 \cdot \vec{p}_2) + V(r)$$

where $(\vec{p}_1, \vec{r}_1)$ and $(\vec{p}_2, \vec{r}_2)$ are the momentum and coordinate of the two particles, respectively, and $r = |\vec{r}| = |\vec{r}_1 - \vec{r}_2|$.

(a) Using the spherical coordinate of the center-of-mass (CM) coordinate, convert the Hamiltonian to

$$H = \frac{1}{2M}(\vec{P}_{CM} \cdot \vec{P}_{CM}) + \frac{1}{2\mu}\left(p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2\sin^2\theta}\right) + V(r)$$

where $M = m_1 + m_2$, $\mu = m_1 m_2/(m_1 + m_2)$, $\vec{P}_{CM}$ is the momentum of the CM, and (p_r, p_θ, p_ϕ) is canonical momentum conjugate to (r, θ, ϕ) that is the spherical coordinate for $\vec{r} = \vec{r}_1 - \vec{r}_2$, respectively.

(b) Show that this two-particles system of 6 degrees of freedom can be converted in to a single-particle system of 3 degrees of freedom with a Hamiltonian of

$$H = \frac{1}{2\mu} p_r^2 + U(r) \qquad \text{where} \quad U(r) = \frac{p_\theta^2}{2\mu r^2} + V(r)$$

(c) For an incident particle with initial speed v_0 when it is very far away from the target particle and with an impact parameter b, calculate the angular momentum p_θ in terms of v_0 and b.

(d) Obtain three integral formula for the trajectory of the scattering

$$\begin{aligned}
t - t_0 &= \sqrt{\frac{\mu}{2}} \int_{r_0}^{r} \frac{dr}{\sqrt{E - U(r)}} && \Longrightarrow && r = r(t) \\
\theta(t) - \theta_0 &= \frac{p_\theta}{\mu} \int_{t_0}^{t} \frac{dt}{r^2(t)} && \Longrightarrow && \theta = \theta(t) \\
\theta(r) - \theta_0 &= \frac{p_\theta}{\sqrt{2\mu}} \int_{t_0}^{t} \frac{dr}{r^2 \sqrt{E - U(r)}} && \Longrightarrow && r = r(\theta)
\end{aligned}$$

where $r_0 = r(t_0)$ and $\theta_0 = \theta(t_0)$.

(e) Assuming you have done the three integration in part **(d)** and have three functions of $r = r(t)$, $\theta = \theta(t)$, and $r = r(\theta)$, explain how you calculate the radiation spectrum of the Bremsstrahlung with arbitrary scattering angle numerically.

(e1) Obtain the formula for a numerical calculation of the velocity and acceleration as a function of t for the scattering.

(e2) Obtain the formula for a numerical calculation of $\vec{A}(\omega)$

(e3) Explain in detail how to calculate the total radiation spectrum $dE/d\Omega$ for arbitrary scattering angle numerically.

(e4) Explain in detail how to calculate the average total radiation spectrum $dE/d\Omega$ over a uniform incident beam scattered inside a matter numerically.

Assignment 15-1 Extra-1

Assignment 15-2 15.2, 15.4